

CS502 Database Management Systems Comprehensive Guide

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Unit 1: ER Modeling and Relational Model

Entity-Relationship (ER) Modeling:

- Entity: An object that exists and is distinguishable from other objects (e.g. Student, Course).
- Attributes: Key attribute (underlined), Composite attribute, Multivalued attribute (double oval), Derived attribute (dashed oval).
- Relationships: One-to-One (1:1), One-to-Many (1:N), Many-to-Many (M:N).
- Mapping ER to Relational Tables: Weak entity sets require identifying relationships, incorporating the primary key of the owner entity.

Unit 2: Functional Dependencies & Normalization

Functional Dependency (FD): $X \rightarrow Y$ holds if whenever two tuples agree on X, they must also agree on Y.

Armstrong's Axioms:

1. Reflexivity: If $Y \subseteq X$, then $X \rightarrow Y$.
2. Augmentation: If $X \rightarrow Y$, then $XZ \rightarrow YZ$.
3. Transitivity: If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$.

Normal Forms:

- 1NF: All attributes must be atomic (no multivalued or composite attributes).
- 2NF: Must be in 1NF and no non-prime attribute is partially dependent on any candidate key (eliminate partial dependencies).
- 3NF: Must be in 2NF and for every non-trivial FD $X \rightarrow Y$, either X is a superkey OR Y is a prime attribute (eliminate transitive dependencies).
- BCNF: For every non-trivial FD $X \rightarrow Y$, X MUST be a superkey.

Unit 3: Transaction Processing & ACID Properties

A transaction is a logical unit of database processing.

ACID Properties:

- Atomicity: All or nothing execution, guaranteed by the Transaction Recovery Manager.
- Consistency: Preserves database invariants before and after commit.
- Isolation: Concurrent transactions execute as if running in isolation, managed by Concurrency Control.
- Durability: Once committed, updates persist even across system crashes, guaranteed by WAL (Write-Ahead Logging).

Unit 4: Concurrency Control & Serializability

Conflict Serializability:

Two operations conflict if they belong to different transactions, access the same data item, and at least one is a write.

Precedence Graph (Serialization Graph):

Draw directed edge $T_i \rightarrow T_j$ if an operation in T_i precedes and conflicts with an operation in T_j . If the precedence graph is acyclic, the schedule is conflict serializable.

Two-Phase Locking (2PL):

- Growing Phase: Transaction acquires locks, cannot release any.
- Shrinking Phase: Transaction releases locks, cannot acquire new ones. Strict 2PL prevents cascading rollbacks.

